

Cryptographic Attribution & Immutable Decryption Provenance

for Multi-Recipient Encrypted Document Distribution

PROBLEM STATEMENT ID	26237
THEME	Blockchain & Cybersecurity
CATEGORY	Software
ORGANIZATION	Ministry of Defence — Indian Navy (WESEE)

TEAM DETAILS

Team Name	[Team Name]	Mentor: [Mentor Name, if assigned]
Team Leader	[Leader Name] — [Roll No.] — [Branch]	
Team Members	1. [Member 2 Name] — [Roll No.] — [Branch]	
	2. [Member 3 Name] — [Roll No.] — [Branch] (Female)	
	3. [Member 4 Name] — [Roll No.] — [Branch]	
	4. [Member 5 Name] — [Roll No.] — [Branch]	
	5. [Member 6 Name] — [Roll No.] — [Branch]	

Proposed Solution & Innovation

CORE IDEA

A per-recipient invisible forensic watermark is embedded at the exact moment of decryption, then bound to that recipient's identity via a post-quantum digital signature and committed to a tamper-evident offline ledger — turning an identical decrypted file into a uniquely traceable one.

USP

Unlike static watermarks (identical across recipients) or access logs (editable by admins), attribution here is session-unique, cryptographically signed by the recipient's own PQC key, and immutably ledgered — fully offline, air-gapped, with zero cloud/KMS or public-chain dependency.

Key Features



Session-Unique Watermarking

Invisible fingerprint generated fresh per decryption — every copy looks identical, forensically distinct.



PQC-Bound Non-Repudiation

ML-DSA signature ties each decryption to the recipient's private key — denial is not possible.



Immutable Offline Ledger

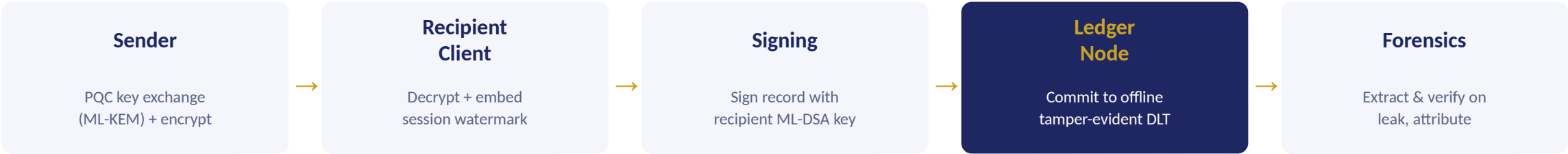
Permissioned DLT records every decryption event; no single admin can alter or erase history.



Leak-to-Recipient Tracing

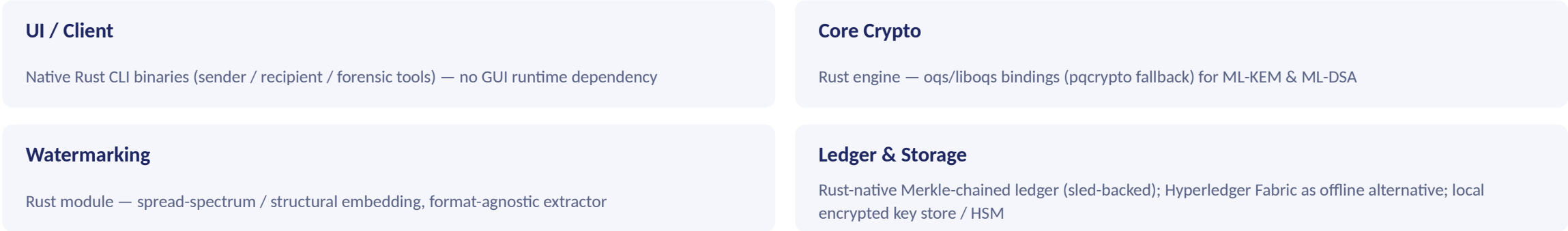
Extract watermark from a leaked copy, match against the ledger, get a verifiable culprit record.

Technical Approach & Architecture

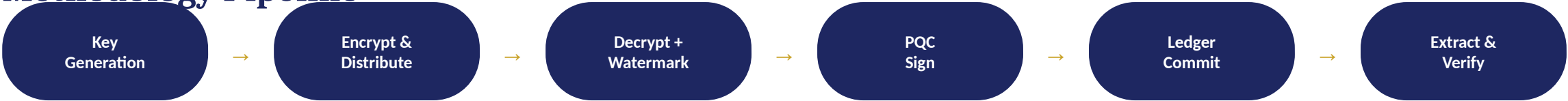


End-to-end data flow: encryption → decryption-time watermarking → PQC signing → ledger commit → leak attribution

Technology Stack



Methodology Pipeline



Steps 3–4 occur locally on the recipient's device before any network/ledger interaction — preserving air-gapped operation.

Feasibility, Risks & Mitigation

TECHNICAL & OPERATIONAL FEASIBILITY

NIST-standardized ML-KEM / ML-DSA are production-ready via the open-source liboqs library; permissioned ledgers (Hyperledger Fabric) already run in fully offline, air-gapped defense networks; steganographic watermarking is an established, well-studied technique — the novelty is composing them into one attribution pipeline.

Potential Challenges

Watermark robustness

Survival across format conversion, screenshots, printing, or re-compression.

PQC performance overhead

ML-KEM/ML-DSA ops add latency on constrained, air-gapped hardware.

Ledger sync without network

Multi-site offline deployments can't broadcast in real time.

Key custody & revocation

No cloud KMS to manage recipient key lifecycle.

Mitigation Strategy

Layered embedding

Combine structural + pixel-domain redundancy; cross-check multiple extraction methods.

Optimized crypto libs

Use liboqs/pqcrypto AVX2-accelerated Rust builds; cache session keys; batch signature verification.

Sneakernet + Merkle proofs

Scheduled manual/removable-media sync with cryptographic Merkle-proof reconciliation.

Local HSM / smartcard

Offline hardware-backed key storage with manually distributed revocation lists.

Impact, Benefits & Scalability

Target Beneficiaries



Measurable Impact

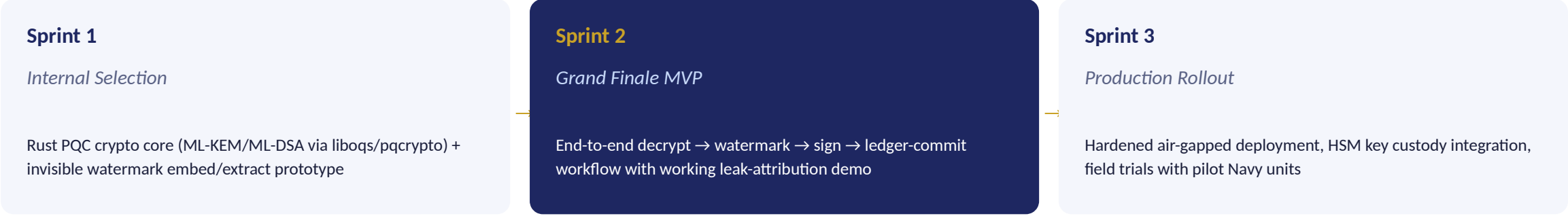


Sustainability & Scalability

Fully offline architecture eliminates recurring cloud costs; NIST post-quantum primitives keep the system secure well beyond the near-term quantum threat horizon; the permissioned ledger scales horizontally across additional secure sites and recipient groups without redesign.

Roadmap & References

Milestone Timeline



References

- NIST FIPS 203 — Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM)
- NIST FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA)
- Open Quantum Safe Project — liboqs reference implementation (Rust bindings via the oqs crate)
- Hyperledger Fabric documentation — permissioned DLT architecture
- Peer-reviewed literature on digital watermarking for document forensics & attribution

Submission Checklist

6-slide limit respected ✓ Visual-first layout ✓ PDF export < 10 MB ✓